IDEA2IMPACT · ONLINE HACKATHON 2026

Theme & Problem Statement

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Theme	Sustainability & Social Impact
Domain	Education & EdTech

1. The Core Vision

Education is the most powerful catalyst for social mobility and long-term societal change. However, the current educational landscape often operates on a "one-size-fits-all" model. Our vision aligns directly with the core idea of building human-first, technology-enabled solutions. By democratizing access to hyper-personalized, 1-on-1 academic support, we aim to level the playing field for students across diverse socioeconomic backgrounds, ensuring that every learner has the tools to overcome their unique academic roadblocks.

2. The Problem Statement

Despite the proliferation of digital learning platforms, a critical pedagogical gap remains: **The Feedback Void.**

When students encounter an error — whether it is a miscalculated mathematics equation, a structurally flawed coding assignment, or a misunderstood physics concept — traditional grading systems only tell them that they are wrong. They rarely explain why they are wrong at a foundational level.

This creates a cascade of systemic issues:

- **The Resource Bottleneck:** In overcrowded classrooms, educators lack the bandwidth to perform a root-cause analysis on every individual mistake made by every student.
- **The Cycle of Frustration:** When students hit a roadblock and cannot identify the specific missing prerequisite knowledge, they experience cognitive overload and disengage from the subject matter entirely.
- **Inequitable Support Systems:** Students with access to private tutors can quickly navigate these roadblocks, while underserved populations are left behind, widening the educational achievement gap.

The fundamental problem is not a lack of educational content, but a lack of targeted, diagnostic intervention at the exact moment a student fails.

3. Our Approach

To solve this, we built a proactive, AI-driven diagnostic engine designed to act as an on-demand, personalized tutor. Rather than passively delivering content, our solution actively analyzes student failures to map and remediate their specific knowledge gaps.

Our technological approach is broken down into four key pillars:

Frictionless Visual Diagnosis

We eliminate the barrier to entry by allowing students to simply upload a document or capture an image of their screen, homework, or worksheet using their device's camera. The system instantly ingests the raw, unstructured data of their mistake.

Deep Root-Cause Analysis (AI Engine)

Utilizing advanced multimodal AI, the engine analyzes the submission to identify the exact “Weak Spot.” It goes beyond surface-level syntax or calculation errors to extract the core concept the student is misunderstanding (the prerequisites).

Targeted Remediation & Interactive Tutoring

The platform immediately provides a custom-tailored “Immediate Fix” alongside curated external resource links. Furthermore, a context-aware AI tutor is instantiated, allowing the student to have an interactive, conversational dialogue to ask follow-up questions about their specific mistake.

Active Knowledge Reinforcement (Dynamic Quizzing)

Passive reading is insufficient for mastery. Our approach closes the learning loop by dynamically generating a targeted, multi-question quiz based exclusively on the identified weak spot. This forces active recall and verifies that the student has truly internalized the core concept before moving forward.

By transforming moments of failure into highly personalized, immediate learning opportunities, our platform uses technology as an enabler to provide equitable, high-quality academic support to any student, anywhere.